Paper 3 Technical Supplement

Admissible Dynamics, Entropy, and the Arrow of Time: Formal Proofs

Companion to Paper 3 of the Admissibility Physics Framework

E. S. Brooke

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Independent Researcher, San Anselmo, California, USA
Corresponding author: brooke.ethan@gmail.com
ORCID: 0009-0001-2261-4682

Abstract

This Technical Supplement provides proof-dense formal treatment of the main theorems of Paper 3 — *Ledgers: Admissible Dynamics, Entropy, and the Arrow of Time.* The main paper states the theorems with proof sketches; the Supplement carries the full formal burden. The organization mirrors the Paper 1, Paper 2, and Paper 7 Technical Supplements. Six formal sections cover, in order: complete positivity as a theorem (T_{CPTP}) in explicit comparison to the Stinespring–Choi–GKSL infrastructure [1, 2, 3, 4, 5]; the directed-enforcement multiplier with $\kappa = 2$ derived from irreversibility and locality (T_κ); von Neumann entropy as committed capacity (T_{entropy}) anchored to the strong-subadditivity backbone [6, 8]; the second law and the arrow of time as structural consequences of L_{irr}; admissibility carrying capacity (ACC) and horizon entropy $S = A/4$ with proof of the $1/2 \times 1/2$ bookkeeping; and the Fisher-information metric on admissible states with RG flow as Fisher gradient descent [12, 13, 14]. A red-team section flags three additional hypotheses at point of use (H1 Markovian-composition closure; H2 Hamiltonian-constraint redundancy; H3 constraint-functional monotonicity). Theorem index and dependency-diagram appendices close the document.

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1. Front matter: inventory of inputs and constitutive principles

1.1 PLEC components imported from Paper 1

Paper 3 imports the four structurally necessary components of PLEC from Paper 1 v4.0-PLEC [16]. They are used throughout this Supplement in the following roles:

Inputs, Constitutive Principles, and Honest Flags

PLEC components (imported).

- **A1** (finite enforcement capacity, upper bound): $\sum_{d \in G} \varepsilon(d) \leq C(\rho, \Gamma)$. Used in every section.
- **MD** (positive cost floor, lower bound): $\inf_{d \in G} \varepsilon(d) \geq \mu^* > 0$. Used in T_κ, T_{entropy}.
- **A2** (argmin selection): $G_{\text{realized}} =_{G \in \mathcal{A}} \sum \varepsilon$. Used in entropy monotonicity (second law) and Fisher-metric uniqueness.
- **BW** (non-degeneracy): cost spectrum rich enough for to be informative. Used in T_{seq} temporal bound.

Structural lemmas (imported). T_{tensor}, L_{loc}, L_{nc}, L_{Δ}, T_{adj} (all from Paper 1); L_{count} ($C_{\text{total}} = 61$, $d_{\text{eff}} = 102$), T_{7B} (gauge-matter sector parity), T_{holonomy}, $B1'$ (real/pseudoreal composite structure) (all from Paper 2).

Additional hypotheses introduced in this paper (flagged at point of use):

- **H1** — Markovian-composition closure. Composition of admissibility-preserving CPTP maps remains admissible. Used in T_{seq} (§2).
- **H2** — Hamiltonian-constraint redundancy. At a causal horizon, the Hamiltonian constraint identifies a gauge-redundant factor of 1/2 in the ACC-to-entropy conversion. Used in the $1/2 \times 1/2$ proof (§6).
- **H3** — Constraint-functional monotonicity. Along admissible trajectories, $\Phi_c = \ln(d_{\text{field}}/d_{\text{admissible}})$ is monotone non-increasing. Used in the staged-cosmogenesis arrow (§5 closing).

None of H1–H3 is a PLEC component. Each is a structural hypothesis local to Paper 3; failures of any would not dissolve the framework but would isolate the corresponding section.

1.2 What is assumed vs. what is proved

Assumed (imported)	Proved in Paper 3	Deferred
A1, MD, A2, BW (PLEC)	T_{CPTP}	Λ_{QCD} (Paper 2)
T_{tensor}, L_{loc} (Paper 1)	T_{κ} ($\kappa = 2$)	PMNS numerics (Paper 4)
L_{nc}, L_{Δ}, T_{adj} (Paper 1)	T_{entropy}	Quantitative cosmogenesis (Paper 6)
L_{count}, T_{7B} (Paper 2)	$T_{\text{second law}}$	Higher-order horizon corrections (Paper 6/7)
H1 (Markov closure)	T_{seq} (temporal Tsirelson)	Planck-scale closure (Paper 5)
H2 (Hamiltonian redundancy)	Horizon $S = A/4$	Expansion history (Paper 6)
H3 (Φ_c monotonicity)	Fisher metric uniqueness	Matter-antimatter asymmetry (Paper 7)

The "proved" column is what this Supplement establishes. The "assumed" column is audited in Papers 1 and 2 (and in the Paper 1 essentiality proofs [16] for the four PLEC components). The "deferred" column is explicitly out of scope for Paper 3.

2. T_{CPTP}: complete positivity as a theorem

The main paper (§4) presents the three-step proof; this section fills in the proofs that were compressed and compares the result explicitly with the Stinespring–Choi–GKSL infrastructure.

2.1 Theorem statement

Theorem 2.1 (T_{CPTP})

Let $\Phi : \mathcal{B}(\mathcal{H}_S) \to \mathcal{B}(\mathcal{H}_S)$ preserve admissibility of the enforcement regime at interface Γ_S. Then Φ is completely positive and trace-preserving (CPTP), and admits a Kraus decomposition $\Phi(\rho) = \sum_k K_k \rho K_k^{\dagger}$ with $\sum_k K_k^{\dagger} K_k = \mathbb{1}$.

2.2 Full proof

Step 1 — Trace preservation. A1 fixes $C(\Gamma_S)$ as a finite positive constant. The normalization $\operatorname{tr}\rho = 1$ is $C_{\text{active}}/C(\Gamma_S) = 1$. Suppose $\operatorname{tr}\Phi(\rho) > 1$. Then committed capacity exceeds $C(\Gamma_S)$, violating A1's upper bound. Suppose $\operatorname{tr}\Phi(\rho) < 1$. Then committed capacity falls below the fixed value; A1 requires both an upper bound and conservation of the budget (because the budget is a constant of the interface, not merely a ceiling). Hence $\operatorname{tr}\Phi(\rho) = 1$.

Step 2 — Complete positivity via ancilla-stability. By T_{tensor} (Paper 1) [16], for any ancilla R the joint system $S \otimes R$ carries an independent budget $C(\Gamma_R)$. The state $\rho_{SR} \geq 0$ is admissible iff all costs are non-negative. Consider $\Phi \otimes \mathbb{1}_R$. At Γ_R, the identity map acts; costs are unchanged; admissibility at Γ_R is preserved trivially. At Γ_S, Φ preserves admissibility by hypothesis. By L_{loc} (Paper 1), the two interfaces are structurally independent: evolution at one cannot create inadmissible states at the other. Hence the joint map is positive on $\rho_{SR} \geq 0$ — i.e., Φ is completely positive.

Step 3 — Kraus representation. Complete positivity + trace preservation on a finite-dimensional Hilbert space (bounded by Paper 1's T_{form}/OR3) invokes Stinespring's dilation [1]: there exists $\mathcal{H}_E$ and an isometry $V : \mathcal{H}_S \to \mathcal{H}_S \otimes \mathcal{H}_E$ with $\Phi(\rho) = \operatorname{tr}_E[V\rho V^\dagger]$. Choi's theorem [2] gives the equivalent Kraus decomposition $\Phi(\rho) = \sum_k K_k \rho K_k^\dagger$ with $K_k = (\mathbb{1}_S \otimes \langle k|_E)V$. The trace-preserving condition translates to $\sum_k K_k^\dagger K_k = \mathbb{1}$. □

2.3 Comparison to Stinespring–Choi–GKSL

A reviewer may reasonably ask: "what does T_{CPTP} add to the already-established Stinespring–Choi–GKSL theory?" The answer is three specific things.

1. **Derivation direction.** In conventional quantum information, the CPTP condition is *imposed* as the definition of a valid evolution map, and Stinespring's theorem [1] then shows that every CPTP map admits a dilation. In APF, the CPTP condition is *derived* from admissibility preservation under A1 and T_{tensor}, and Stinespring's theorem then supplies the Kraus form. The direction of the derivation is reversed.

2. **Physical interpretation of the dilation.** Stinespring's environment $\mathcal{H}_E$ is traditionally treated as a mathematical device. In APF, $\mathcal{H}_E$ is a second enforcement interface with its own capacity budget $C(\Gamma_E)$; the partial trace corresponds to abandoning the Γ_E ledger. The unitary evolution on $\mathcal{H}_S \otimes \mathcal{H}_E$ is the admissibility-preserving joint dynamics of two coupled budgets.

3. **Restriction to finite dimension.** Paper 1's T_{form} bounds $\dim \mathcal{H}_S \leq \lfloor C/\mu^* \rfloor$ at each interface. Stinespring's theorem applies in full generality (type I_∞ and beyond); APF restricts to type I_n with $n < \infty$. The GKSL–Lindblad generators [3, 4] then live in $\mathcal{B}(\mathcal{H}_S)$ with explicit finite trace.

These three differences are what makes "CPTP as a theorem" a nontrivial claim rather than a change of language. The underlying mathematics is unchanged; the provenance is.

2.4 Sequential composition and the temporal Tsirelson bound

Theorem 2.2 (T_{seq})
Under H1 (Markovian-composition closure), any T-step sequence of admissible CPTP maps $\Phi_T \circ \cdots \circ \Phi_1$ is itself admissible. Time-evolved dichotomic observables $a(t) = U_t^\dagger a U_t$ satisfy the Cirel'son operator identity; the temporal CHSH correlator is bounded by $2\sqrt{2}$.

Proof. CPTP maps form a semigroup under composition; the composition of trace-preserving maps is trace-preserving; the tensor product of two Kraus decompositions is a Kraus decomposition for the composed map. This is standard [3, 4]; H1 is the requirement that admissibility *preservation* commutes with this composition (admissibility-preserving at each step implies admissibility-preserving for the composite). For the temporal CHSH bound: the Cirel'son identity $\|[a(t_1), a(t_2)]\|_{\text{op}} \leq 2$ uses only the algebraic structure of self-adjoint unitaries (via T_{adj}, Paper 1) and is state-independent, giving $|\omega(\hat{S}_T)| \leq 2\sqrt{2}$ for the operator $\hat{S}_T = a(t_1) \otimes (b(t_1) + b(t_2)) + a(t_2) \otimes (b(t_1) - b(t_2))$. □

Code reference: `check_T_seq` in `core.py`.

3. T_κ: the directed-enforcement multiplier $\kappa = 2$

3.1 Theorem statement

Theorem 3.1 (T_κ)
In the directed-enforcement regime (one interface committed to enforcing a distinction against a second interface), the per-distinction enforcement multiplier satisfies $\kappa = 2$ uniquely. Values $\kappa < 2$ correspond to erasable single-interface commitment (not directed); values $\kappa > 2$ require an additional independent correlation that is not cost on the original distinction.

3.2 Derivation of $\kappa = 2$

The multiplier κ answers: how much capacity is committed per distinction when enforcing a directed (irreversible) commitment? The minimum at each interface is μ^* (MD); the directed enforcement commits at each of two interfaces (the distinction-holder and the distinction-witness).

Upper bound $\kappa \leq 2$. A directed distinction requires commitment at two interfaces only: the distinction-holding interface (committing μ^* to maintain the distinction) and the witnessing interface (committing μ^* to record the distinction). A third interface commitment would be an independent correlation — a separate distinction in its own right, not a higher κ on the original. By Paper 1 L_{nc}, multiple simultaneous commitments do not compress into a single distinction's cost; they are separate entries in the ledger. Hence $\kappa \leq 2$.

Lower bound $\kappa \geq 2$. If $\kappa < 2$, the directed distinction is enforceable at less than $2\mu^*$. Two sub-cases: $\kappa = 0$ is excluded by MD (no zero-cost distinctions); $0 < \kappa < 2$ would place the commitment below the floor of two interfaces, which by L_{irr} contradicts the irreversibility of directed commitment (single-interface commitment is erasable; directed commitment is not). Hence $\kappa \geq 2$.

Uniqueness. Both bounds saturate at $\kappa = 2$. □

Code reference: `check_T_kappa` in `core.py`.

3.3 Numerical verification

The codebase evaluates κ on the 61-mode ledger and returns $\kappa = 2$ with no approximation: the computation is purely combinatorial over the $C_{\text{total}} = 61$ capacity types and yields $\kappa = 2$ identically (see *Code reference:* `check_T_kappa` in `core.py`.). This is *not* a numerical coincidence: the derivation is structural, and the code merely corroborates what the proof requires.

4. T_{entropy}: von Neumann entropy as committed capacity

4.1 Theorem statement

Theorem 4.1 (T_{entropy})
The von Neumann entropy $S(\rho) = -\text{tr}(\rho \log \rho)$ equals κ^{-1} times the enforcement demand of the active correlations at the system interface:

$$S(\rho) = \frac{1}{\kappa}\,\mathcal{E}_{\text{cross}}(\rho) = \frac{1}{2}\,\mathcal{E}_{\text{cross}}(\rho),$$

where $\mathcal{E}_{\text{cross}}$ counts the cross-interface enforcement cost (in nats) committed by ρ. The standard entropy properties (positivity, concavity, subadditivity, strong subadditivity, data-processing inequality) follow from A1, L_Δ, T_{CPTP}, and Lieb–Ruskai [6].

4.2 Derivation

Step 1 (Identification). By Paper 1 T_{adj}, the cost functional on the operator algebra $\mathcal{A}$ induces an inner product; the cost of a projector P_i is $\varepsilon(P_i) = \text{tr}(\rho P_i)$ for a state ρ. For a rank-one projector $P_i = |\psi_i\rangle\langle\psi_i|$ in the eigenbasis of ρ with eigenvalue λ_i, $\varepsilon(P_i) = \lambda_i$. The total cross-interface enforcement demand is $\mathcal{E}_{\text{cross}}(\rho) = -2\sum_i \lambda_i \log \lambda_i$ (the factor of $\kappa = 2$ comes from directed commitment at both the system and the environment interfaces; see T_κ). Hence $S(\rho) = \mathcal{E}_{\text{cross}}/2 = -\sum_i \lambda_i \log \lambda_i$, which is the von Neumann entropy.

Step 2 (Standard properties from imports).

- **Positivity.** $S(\rho) \geq 0$ from $\lambda_i \in [0, 1]$.
- **Concavity.** Follows from the concavity of $-x \log x$; preserved by cost-functional linearity on the algebra.
- **Subadditivity.** $S(\rho_{SE}) \leq S(\rho_S) + S(\rho_E)$ follows from L_Δ applied to co-located distinctions at $\Gamma_S \cup \Gamma_E$.
- **Strong subadditivity.** $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$ is Lieb–Ruskai [6]; it follows in APF from the admissibility of the Naimark dilation of the tripartite structure and T_{CPTP}.
- **Data processing.** Under any CPTP map Φ, the relative entropy is non-increasing [8]. In APF, this is T_{CPTP} plus L_{irr}: committed cross-interface correlations cannot decrease under local operations.

□

Code reference: `check_T_entropy` in `core.py`.

4.3 Comparison to Lindblad and Spohn

The mainstream open-systems route to the second law runs through Lindblad's CP-map entropy inequalities [3] and Spohn's positive-entropy-production framework for quantum dynamical semigroups [7]. Spohn established that the entropy production $\sigma(\rho) := -\frac{d}{dt}D(\rho_t\|\rho_*)$ is non-negative and convex along the semigroup flow, where ρ_* is the stationary state and D is the relative entropy; this gives approach to equilibrium for broad classes of open quantum dynamics.

The framework here is in alignment, not in competition. Three differences of presentation, none of mathematics:

1. **Source of irreversibility.** Spohn locates irreversibility in the semigroup structure of the open-system dynamics. The framework here locates it in L_{irr}: local unrecoverability of cross-interface correlations. These are two different routes to the same monotonicity statement.

2. **Normalization.** Spohn's entropy production is naturally formulated relative to the stationary state ρ_*. The framework here works directly with the von Neumann entropy of ρ_t, with the factor of $\kappa = 2$ supplying the directed-enforcement multiplier that the open-systems literature usually absorbs into the choice of generator.

3. **Scope.** Spohn's framework is deeply developed for Markovian semigroups; the admissibility framework's T_{seq} (§2) extends to multi-step composition under hypothesis H1 (Markovian-composition closure), explicitly flagged as a separable hypothesis whose failure would isolate the multi-step second-law statement without affecting the single-step one.

Paper 3's contribution to entropy theory is rhetorical and structural, not a new monotonicity inequality. The Lieb–Ruskai backbone is unchanged; Spohn's entropy-production positivity is unchanged. What changes is the account of *why* the monotonicity holds.

4.4 The ledger-accounting principle (formal statement)

Proposition 4.2 (Ledger-Accounting Principle)
The entropy count is invariant under small changes in the enforcement-cost landscape. Formally: let $\varepsilon'(d) = \varepsilon(d) + \delta(d)$ be a perturbation of the cost functional with $\|\delta\|_\infty < \mu^*$. Then the count of admissible configurations at fixed capacity $C(\Gamma)$ is unchanged: the admissible set under ε' differs from that under ε only at the boundary, and the entropy functional $S(\rho) = \ln N_{\text{admissible}}$ is invariant up to $O(\delta/\mu^*)$ corrections that vanish as the perturbation vanishes.

Sketch. The admissible set $\mathcal{A}$ is defined by two inequalities: $\inf \varepsilon(d) \geq \mu^*$ and $\sum \varepsilon(d) \leq C$. Perturbations with $\|\delta\|_\infty < \mu^*$ preserve the floor condition strictly; the budget condition shifts by $\sum \delta(d) = O(|G|\|\delta\|_\infty)$, which for bounded cost spectra is bounded. The count $|\mathcal{A}|$ is invariant up to boundary fluctuations of this order. Entropy as $\ln|\mathcal{A}|$ inherits the invariance. □

Remark. This is the formal statement of the claim used in the main paper (§6.3, ledger-accounting principle) that entropy is robust under cost-landscape corrections. Paper 7's partition-function treatment [19] elevates this to a full thermodynamic argument; here it functions as a direct count-based observation.

5. $T_{\text{second law}}$ and the arrow of time

5.1 L_{irr} (formal statement and proof)

Lemma 5.1 (L_{irr})
Let S and E be systems at independent interfaces Γ_S and Γ_E (by L_{loc}). After an interaction producing cross-interface correlation $\Delta(S, E) > 0$ (by L_Δ), the pre-interaction reduced state at Γ_S is unrecoverable by any CPTP map local to Γ_S.

Proof. Suppose for contradiction that Φ_S is a Γ_S-local CPTP map reversing the interaction as seen from Γ_S: $\Phi_S(\rho_S^{\text{after}}) = \rho_S^{\text{before}}$. The cross-interface capacity committed by the correlation is $\Delta > 0$; this capacity is committed simultaneously at Γ_S and Γ_E (by L_Δ). For Φ_S to reclaim the full pre-interaction state, it would need access to the Γ_E-side commitment. By L_{loc}, Φ_S has no such access. The recovered reduced state at Γ_S therefore differs from ρ_S^{before} by the Γ_E-side capacity commitment, contradiction. □

Code reference: `check_L_irr` in `core.py`.

5.2 $T_{\text{second law}}$ (formal statement)

Theorem 5.2 ($T_{\text{second law}}$)
Along any admissible sequence of CPTP transitions $\rho_0 \to \rho_1 \to \cdots \to \rho_T$, the von Neumann entropy is non-decreasing:

$$S(\rho_0) \leq S(\rho_1) \leq \cdots \leq S(\rho_T).$$

Equality $S(\rho_t) = S(\rho_{t+1})$ holds only when the t-th transition commits no new cross-interface correlation.

Proof. At each step, the transition $\rho_t \to \rho_{t+1}$ is either (a) unitary (admits an inverse CPTP; equality case), or (b) involves cross-interface coupling ($\Delta > 0$). In case (b), L_{irr} guarantees the pre-transition state is unrecoverable; the data-processing inequality [6, 8] on the coupled system $\rho_{SE,t}$ then forces $S(\rho_{S,t+1}) \geq S(\rho_{S,t})$. Summing over steps gives the chain. □

Code reference: `check_T_second_law` in `core.py`.

5.3 The arrow of time and T-violation at $\pi/4$

Theorem 5.3 (T_{CPT})
The structural T-violation phase is $\theta_T = \pi/4$. This value follows from L_{irr}, the holonomy-phase lemma $L_{\text{holonomy phase}}$ (Paper 2), and the algebraic structure of Paper 1's T2 (finite-dimensional complex Hilbert-space representation).

Sketch. The holonomy argument of Paper 2 [17] shows that directed admissible cycles in the state space pick up phases quantized in units related to the $\mathbb{Z}_2 \times \mathbb{Z}_2$ structure of the gauge sector. The minimum directed-cycle T-violation phase is half of $\pi/2$, i.e. $\pi/4$ — the result of one L_{irr}

step integrated around a smallest admissible directed loop. □

Code reference: `check_T_CPT` in `core.py`.

5.4 Macroscopic arrow from loss of global coordination

Proposition 5.4 (Macroscopic arrow)
At cosmological scale, a globally coordinated admissible state is one in which admissibility holds jointly across all interfaces in a causal region. Under H3 (monotonicity of the constraint functional Φ_c), the set of globally coordinated states is strictly decreasing along the admissible trajectory: coordination can be lost but not regained. The arrow of time is the direction of coordination loss.

This is L_{irr} applied to the coordinated-admissibility observable over a macroscopic region; the proof is the same structure as L_{irr}, with "local interface" replaced by "local causal region."

6. Admissibility carrying capacity and $S = A/4$

6.1 ACC: formal definition

Definition 6.1 (Admissibility Carrying Capacity)**.** *Let Γ be a relational interface (a causal boundary, a quantum bipartition, or any co-dimension-1 surface across which admissibility is enforced). Its admissibility carrying capacity is $C_{\text{ACC}}(\Gamma)$ = the supremum number of distinct admissible configurations that can coexist at the interface, weighted by their cost in the enforcement currency.*

6.2 The $1/2 \times 1/2$ bookkeeping

Theorem 6.1 (Horizon entropy prefactor)
At a causal horizon with area A, the entropy is

$$S_H = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{A}{\ell_P^2} \cdot \frac{\ln d_{\text{eff}}}{\ln 2} = \frac{A}{4\ell_P^2} \cdot \frac{\ln d_{\text{eff}}}{\ln 2},$$

which, after the conventional ACC-to-bit conversion, is $S_H = A/4$ in natural units — the Bekenstein–Hawking area law [9, 10].

Proof. Two structurally forced factors of 1/2 compose.

Factor 1 (Relational symmetrization). The ACC counts relational distinctions across the interface. Each distinction is a pair (a, b) where a is on one side and b is on the other; under relational equivalence $(a, b) \sim (b, a)$, each pair is counted twice in the naive bookkeeping. Symmetrization gives a factor of 1/2.

Factor 2 (Hamiltonian-constraint redundancy; hypothesis H2). At a causal horizon, general covariance imposes the Hamiltonian constraint: time reparameterizations at the horizon are gauge. This identifies two admissible configurations differing only in their gauge slicing as physically identical, giving a second factor of 1/2. H2 is the assumption that this identification

is the *only* relevant gauge redundancy; other gauge symmetries (spatial diffeomorphisms, internal gauge) are absorbed into the interface specification upstream.

Multiplying: $1/2 \times 1/2 = 1/4$, which is the Bekenstein–Hawking prefactor. □

Code reference: `check_T_deSitter_entropy` in `gravity.py`.

6.3 Unification of three entropies

Corollary 6.2 (Three-entropies unification)**.** *Thermodynamic entropy, entanglement entropy, and horizon entropy are three specializations of the same ACC-saturation count. They differ only in what the interface Γ is:*

- *Thermodynamic: Γ is the system/environment boundary (implicit coarse-graining).*
- *Entanglement: Γ is the quantum bipartition.*
- *Horizon: Γ is the causal boundary.*

Remark. This unification is a clear statement that the framework subsumes rather than supplants standard black-hole and cosmological-horizon thermodynamics. The conventional $S = A/4$ law of Bekenstein and Hawking [9, 10] is recovered; the de Sitter cosmological-horizon entropy of Gibbons and Hawking [11] is recovered as the saturated case $S_{dS} = C_{\text{total}} \ln d_{\text{eff}} = 61 \ln 102$, with the combinatorial values inherited from Paper 2's capacity-partitioning argument. The additional claim is that the same ledger underlies the other two regimes. The unification does not contradict any observation; it reorganizes the existing explanations around a single counting principle.

6.4 Comparison to Bekenstein–Hawking and Gibbons–Hawking

The standard derivations of horizon entropy operate at the semiclassical level: Bekenstein's information-theoretic argument from the area law [9], Hawking's particle-creation calculation yielding the thermal spectrum at horizon temperature $T_H = \hbar\kappa_{\text{surf}}/2\pi k_B$ [10], and the Gibbons–Hawking extension to cosmological event horizons in positive-Λ spacetimes with horizon temperature $T_{dS} = H/2\pi$ [11]. Each gives $S = A/4$ in natural units.

The framework here is in the same lineage; the $1/2 \times 1/2$ decomposition is a specific combinatorial account of where the standard 1/4 prefactor comes from, valid under H2 (Hamiltonian-constraint redundancy as the only relevant gauge structure in the horizon's ACC-to-entropy conversion). We do not provide a competing semiclassical derivation, and the result is the same horizon entropy in either language. The framework's specific addition is the ACC vocabulary that makes thermodynamic, entanglement, and horizon entropy three specializations of one accounting (the corollary above); this is a unification of three separate semiclassical results, not a re-derivation of any of them.

7. Fisher information and RG flow

7.1 Fisher metric uniqueness

Lemma 7.1 (L_{Fisher})
The unique Riemannian metric on the space of admissible density matrices that (a) is monotone under CPTP maps, (b) reduces to the classical Fisher metric on commuting states, and (c) is invariant under unitary conjugation, is the quantum Fisher information metric (equivalently, the symmetric logarithmic derivative / Bures metric) of Braunstein–Caves [12]. Under the capacity-counting normalization of Paper 2 ($d_{\text{eff}} = 102$),

$$S = \frac{d_{\text{eff}}}{2} \ln \det G,$$

where G is the Fisher metric on admissible states.

Sketch. Monotonicity under CPTP pins the metric up to a one-parameter family of admissible metrics in the Petz classification [13]. Among these, the symmetric logarithmic derivative is distinguished by (b)+(c), reducing to the classical Fisher metric on commuting states with unitary invariance. The entropy-determinant relation follows from the capacity-counting structure of Paper 2 [17]: the entropy at ACC saturation is $C_{\text{total}} \ln d_{\text{eff}}$, and the volume form of G scales as $\det G$ by standard arguments from information geometry [14]. □

Code reference: `check_L_Fisher_measure` in `core.py`.

7.2 RG flow as Fisher gradient descent

Proposition 7.2 (RG flow as Fisher descent)
Under the Fisher metric G, the renormalization-group flow on the space of admissible effective descriptions is the gradient descent of an entropy functional. Coarse-graining steps (integrating out high-momentum modes) correspond to moves in the direction of maximum entropy gain per capacity used — the gradient of S under G.

Sketch. Coarse-graining is a CPTP map on the effective theory. By the data-processing inequality, entropy is non-decreasing under coarse-graining. Under the Fisher metric, the steepest-ascent direction for entropy is the gradient; stationary points of the RG flow are fixed points of this gradient descent. The traditional RG beta-function interpretation is recovered with the identification of the beta-function with the Fisher gradient [14]. Full dynamical analysis is Paper 4 [18] territory for specific flavor channels.

7.3 Comparison to Bény–Osborne

The closest external literature for the RG-as-Fisher-descent claim is the program of Bény and Osborne [15], who formulate RG explicitly as a family of quantum channels with an induced information geometry on the space of admissible effective descriptions. In their formulation, the RG step is a CPTP map, monotone RG-invariants are information-theoretic quantities non-increasing along the flow, and the Fisher metric provides the natural geometric object for characterizing information loss.

The framework here is directionally aligned: RG as CPTP, Fisher metric as the induced geometry, entropy gradient as the direction of RG flow. The additions specific to the admissibility framework are (i) the capacity-count normalization ($d_{\text{eff}} = 102$) from Paper 2's structural argument and (ii) the connection to the Frobenius–Nagata generation-channel structure of Paper 4. Neither of these is provided by the Bény–Osborne formulation; neither contradicts it. Paper 3's contribution is therefore a specific embedding of APF capacity counting into the Bény–Osborne information-geometric RG picture, not a re-derivation of that picture.

8. Red-team section: additional hypotheses at point of use

Three hypotheses are used in this Supplement that are *not* PLEC components. Each is flagged below at the point where it enters, with the scope of what it supports and what would survive its failure.

H1 — Markovian-composition closure (§2)

Statement. The composition of admissibility-preserving CPTP maps is admissibility-preserving.
Scope. Used in T_{seq} (temporal CHSH bound) and in the second-law chain. Without H1, the second law still holds step-by-step but not as a global statement about multi-step evolutions.
Status. [P] in Paper 7's partition-function framework; [C] in Paper 3 alone. The admissibility class is closed under composition if and only if the composed map is itself admissibility-preserving. Cases where non-Markovian memory could violate this (long-range correlations at a saturating interface) would falsify H1 for specific regimes.
Red-team attack. Construct a pair of admissibility-preserving maps whose composition is not admissibility-preserving. None is known within the framework, but the construction is open as a check on H1.

H2 — Hamiltonian-constraint redundancy (§6)

Statement. At a causal horizon, the only relevant gauge redundancy in the ACC-to-entropy conversion is time reparameterization (Hamiltonian constraint). Other gauge symmetries are absorbed upstream.
Scope. Provides the second factor of 1/2 in the $1/2 \times 1/2$ bookkeeping for horizon entropy. Without H2, the horizon-entropy prefactor would require a different account; the ACC structure itself would survive.
Status. Consistent with general-relativistic canonical analysis (ADM formulation). [C] pending explicit derivation from Paper 6's geometric admissibility framework.
Red-team attack. Show a causal horizon where spatial diffeomorphisms contribute an additional gauge redundancy beyond the interface specification, changing the prefactor. Plausible candidates: rotating black holes (Kerr) with non-trivial angular gauge structure; anti-de Sitter horizons with boundary conditions that modify the counting. Neither is worked out here.

H3 — Constraint-functional monotonicity (§5 closing)

Statement. Along admissible trajectories, the constraint functional $\Phi_c = \ln(d_{\text{field}}/d_{\text{admissible}})$ is monotone non-increasing.

Scope. Underwrites the macroscopic arrow of time and the staged cosmogenesis picture of the main paper (§9 of the main paper). Without H3, the quantum-interface arrow (L_{irr}) still survives; the cosmological-scale arrow would require an alternative argument.

Status. [P+H3] in this paper; [P] under the Paper 6 development of the cost-landscape dynamics for gravitational degrees of freedom.

Red-team attack. Exhibit an admissible cosmological trajectory where Φ_c fails monotonicity — e.g., a bouncing-cosmology model that recovers earlier-epoch admissible configurations. The framework predicts no such trajectory exists.

Appendix A: Theorem Index

Result	Type	Depends on	Paper 3 section
T_{CPTP}	Theorem	A1, T_{tensor}, L_{loc}	§2
T_{seq}	Theorem	T_{CPTP}, T_{adj}, H1	§2
T_κ	Theorem	MD, L_{irr}, L_{nc}	§3
T_{entropy}	Theorem	T_κ, T_{adj}, L_Δ, Lieb–Ruskai [6]	§4
Ledger-accounting Prop.	Proposition	A1, MD	§4
L_{irr}	Lemma	A1, L_{nc}, L_{loc}, L_Δ	§5
$T_{\text{second law}}$	Theorem	L_{irr}, T_{entropy}	§5
T_{CPT} ($\theta_T = \pi/4$)	Theorem	L_{irr}, T_{holonomy}, T2	§5
Macroscopic arrow	Proposition	L_{irr}, H3	§5
Horizon $S = A/4$	Theorem	ACC, H2, Bekenstein [9], Hawking [10]	§6
Three-entropies unification	Corollary	T_{entropy}, ACC	§6
L_{Fisher}	Lemma	Braunstein–Caves [12], Petz [13], T_{CPTP}	§7
RG flow as Fisher descent	Proposition	L_{Fisher}, data processing	§7

Appendix B: Dependency diagram

```
                    A1 + MD + A2 + BW  (PLEC, from Paper 1)
                              |
            +------------+------------+
            |                         |
   T_tensor (Paper 1)         L_nc, L_Delta, L_loc (Paper 1)
            |                         |
            v                         v
      T_CPTP            L_irr  +---> T_kappa (kappa = 2)
            |              |                |
            | H1           |                v
            v              |        T_entropy <- Lieb-Ruskai [LR73]
         T_seq             |                |
                           v                v
                    T_second_law    Ledger-accounting
                              |
                     +-------+-------+
                     |               |
               T_CPT (pi/4)    Macroscopic arrow (H3)
```

```
                                  |
                                  v
                        Cosmogenesis (Paper 3 main)

ACC + H2 -> Horizon S = A/4 -> Three-entropies unification
       |
       |
       +-- L_Fisher -> RG flow as Fisher descent
             |
             v
        (Paper 4 flavor channels)
```

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